

Short Article

on

WATER POLLUTION IN BANGLADESH

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1. Background:

Three essential natural resources-water, soil, and air-are necessary for human subsistence on Earth. Among which water is a liquid of life since life cannot exist without water. “About 71% of the earth's surface is made up of water, which is originate across seas, rivers, lakes, glaciers, and groundwater systems. Water is playing an essential role in shaping ecosystems and supporting life.” [1]

Water pollution occurs when water bodies, including rivers, lakes, seas, and groundwater, are contaminated by harmful substances that degrade water quality and interrupt ecosystems. Both human activity and natural processes are responsible for this pollution, although human activity is the main cause. [2] “Water quality is influenced by multiple variables including precipitation, climate, soil type, vegetation, geology, flow conditions, ground water, and human activities. The paramount threat to water quality is posed by point sources of industries and municipalities. It is also impacted by mining, urbanization, and agriculture. Sediments, fertilizers, and harmful pollutants are also considered non-point sources of pollution.” [3] “Polluted water is a major cause of diseases and deaths worldwide, with an estimated 14,000 deaths each day.” [4]

“One of the world's most densely populated nations, Bangladesh, has an abundance of water sources, yet these are constantly being contaminated. The fast urbanization and industrialization of Bangladesh are polluting both surface and groundwater, and most significantly, industrial waste and arsenic contamination are compromising the potability of this natural resource. Bangladesh has a high rate of water-borne disease-related deaths, especially among children. The major causes of water pollution are anthropogenic sources, including untreated industrial effluents, inappropriate household waste disposal, and agricultural runoff. For minimize the pollution the government approved the laws, ordinances, policies, and actions; however due to lack of proper implementation and monitoring water pollution problem is getting worse every day.” [5] [6] [7]

2. Situation in Bangladesh:

Bangladesh is among the world's most overpopulated countries which is a lower-riparian, riverine, and naturally resourceful country. This country has enough water to sustain itself. [8] “Bangladesh withdraws about 32 km³ of groundwater annually, of which 90% is used for irrigation and the other 10% is used for household and industrial purposes. That amount is

equivalent to 4% of the world's groundwater extraction.” [9] Bangladesh's surface water comes from lakes, ponds, canals, rivers, and streams. Bangladesh shares fifty-four rivers with India and three with Myanmar. All these rivers are helping to build nearly 230 rivers across the country. [8] A Daily Star report claims that in addition to the enormous number of rivers, there are more than 1.3 million ponds and lakes in the whole of the country. (The Daily Star,2016). Water quality of Bangladesh is decreasing day by day. Water supplies in Bangladesh are in danger due to a confluence of factors such as pollution, climate change, and population increase, which can occasionally be brought on by poor management. [9] [10]

“Bangladesh's dense population, fast industrialization, and poor waste management systems all contribute to the country's serious water contamination problems. These problems pose a threat to ecosystems, public health, and the nation's socioeconomic stability because water is an essential resource for daily life, fishing, and agriculture.” [11] Major Causes of Water Pollution in Bangladesh is discussed below:

- › **Arsenic Contamination:** In Bangladesh, 98% of people rely on groundwater as their primary source of drinking water, and there are between 6 and 11 million tube wells. [12] “Bangladesh has experienced one of the most serious public health emergencies in its history which is arsenic poisoning. The problem first surfaced in the 1990s when elevated amounts of arsenic were found in drinking groundwater, especially in rural regions. However, there is a significant risk to those who drink groundwater that has been contaminated with Arsenic (As) due to natural processes and excessive groundwater removal brought on by population development. Anawar et al. claim that a staggering amount of As (>50 µg/L) has significantly polluted 27% of Bangladeshi wells. We know that drinking ground water polluted with arsenic can cause arsenic poisoning in humans, but there are other ways to get it, such as eating food contaminated with arsenic. Thus, eating rice, a major food, exposes the Bangladeshi population to arsenic as well. Due to irrigation with arsenic-contaminated water, paddy soils in Bangladesh have been discovered to possess elevated levels of arsenic.” [13] [14] [15]
- › **Industrial Discharges:** “Bangladesh is encouraging fast industrial growth, but industries also contribute to major environmental issues because many of them release their waste, including potentially hazardous materials, without treatment directly into the environment. Industrial wastewater has a number of unfavorable traits. Bangladesh's primary industries are classified as follows: textile, tannery, fertilizer, paper and pulp, vegetable oil, sugar, cement, steel, paint, and chemicals. The majority of the industry do

not have any type of treatment facilities and frequently release their raw waste, which comprises a variety of dangerous and toxic substances, either straight to a nearby lake, river, or khal or through the local storm drainage system.” [16]



Figure 2.1: Industrial wastewater release directly into the surface water.

The Bangladesh Center for Advanced Studies (BCAS) performed an industrial assessment in 2009 and found that only approximately 40% of industries have ETPs. Around 50% of industries do not establish any ETP facility, while 10% of industries had ETPs in construction at the time. [17] Water supplies are severely contaminated in Dhaka and its neighboring districts, such as Gazipur and Narayanganj, which are the worst affected by unplanned industrialization. The water quality is contaminated by the improper discharge of untreated wastewater from different industries that are close to these rivers. [6] [17]

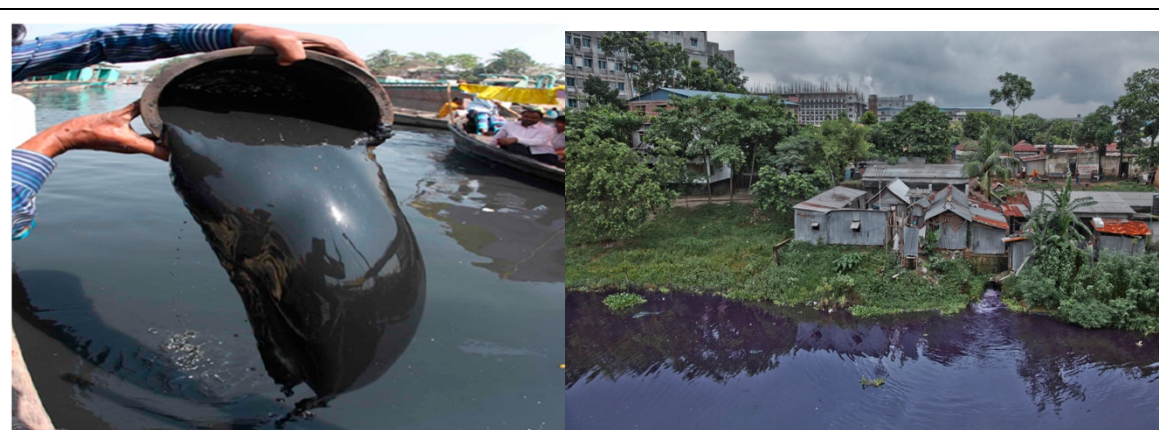


Figure 2.2: Appearance of Buriganga river water in dry season.

- › **Agricultural Reasons:** Agricultural water pollution affects aquatic ecosystems and water quality, with far-reaching effects on the environment, human health, and food security. It is brought on by pollutants from both point and non-point sources, including sediment,

fertilizers, and pesticides. The local farmers use various kinds of carbamate and organophosphorus insecticides extensively, but they abuse them since they lack the necessary training and understanding. Pesticide misuse results in contamination of surface and groundwater, primarily from rainfall or runoff from agricultural areas. [18]

- › **Sewage and Domestic Waste:** “Sewage and domestic waste are significant contributors to water pollution in Bangladesh, especially in urban areas like Dhaka and Chittagong. The country’s current wastewater management system is overburdened by the country’s increasing urbanization and population, resulting in the direct release of untreated sewage into rivers, canals, and other water bodies. The Buriganga, Turag, and Shitalakshya are among the major rivers that are heavily contaminated by domestic sewage.” [19]
- › **Plastic and Solid Waste:** The accumulation of non-biodegradable plastics in water bodies endangers aquatic life. The Buriganga, Turag and Karnaphuli rivers are major locations for the accumulation of solid and plastic garbage. [20]



Figure 2.3: Plastic and Solid Waste from urban area to pollute rivers in Bangladesh.

Impacts of water pollution: “In Bangladesh, the increasing rate of water pollution has become of serious health hazard. Water pollution has a significant impact on living things, including humans and animals, in addition to the environment. Many fatal water-borne illnesses, such as diarrheal disorders, are mostly caused by drinking contaminated water in rural regions. According to the 2004 Bangladesh Demographic and Health Survey (BDHS), diarrhea was a contributing factor in 1.2% of neonatal deaths and 5.1% of fatalities in children under five. According to a different study, diarrhea was the cause of 0% of neonatal fatalities and 6.9% of deaths in children under five. Drinking water contaminated by microorganisms raises the risk of contracting hepatitis A, hepatitis B, diarrhea, dysentery, and typhoid. Fish were primarily found in the country’s rivers a few centuries ago. Heavy metals

including lead, mercury, and arsenic are among the industrial pollutants that contribute to long-term health concerns like cancer, neurological diseases, and developmental abnormalities in children. However, fish populations have been decreasing in the majority of Bangladesh's rivers due to the increasing prevalence of water pollution. As a result of water contamination, several fish species have already gone extinct. Biodiversity is being lost as a result of rivers like the Buriganga and Shitalakshya being too filthy to sustain aquatic life. Pollutants from plastic and chemicals reach the Bay of Bengal, harming marine ecosystems and endangering fish and turtle populations. So, Water contamination has serious negative effects on Bangladesh's economy, society, ecology, and health. [21] [22]

3. Ways to deal:

According to the Bangladesh Country Environmental Analysis (CEA), lead exposure, inadequate sanitation and hygiene, contaminated water, and air pollution result in about 272,000 preventable deaths every year. In 2019, the cost of these environmental problems amounted to 17.6% of Bangladesh's GDP. Bangladesh's water pollution problem necessitates a multifaceted strategy that includes foreign assistance, community engagement, government policy, and technological innovation. For long-term progress, these remedies must address the underlying causes of pollution while encouraging sustainable practices. [23] “This country has also some regulations for the prevention of water pollution and there are so many effective solutions that can help with the reduction of pollution in all water bodies. Now, discuss about some strategies to overcome the water bodies pollution in Bangladesh-

Strengthening municipal wastewater management

- › Develop wastewater treatment plants (WWTPs): Dhaka, Gazipur, and Chattogram are among the cities most badly impacted by surface water pollution, and none of them have municipal wastewater management facilities. So, there should be build and update sewage treatment plants to clean up household and industrial wastewater before it gets thrown into rivers.
- › Industrial wastewater management: Encourage companies to employ Effluent Treatment Plants (ETPs), mainly in tanneries, dyeing facilities, and textiles and implement Cleaner Production Technologies. Tanneries and other highly polluting enterprises should be moved to industrial zones with adequate waste treatment facilities rather than into dense urban cities.

Better Farming Methods use for control water pollution

- › For reducing chemical runoff, Train farmers effective irrigation methods and Integrated Pest Management (IPM) to minimize chemical leaching and encourage the use of biopesticides and organic fertilizers to reduce nutrient contamination in waters.
- › Establish vegetation buffer zones adjacent to areas of water to absorb runoff and replenish wetlands to act as organic pollutants' filters.

Resolving of Arsenic Contamination

- › Encourage the use of surface water purification, rainwater collection, and deep tube wells in areas afflicted by arsenic.
- › Development of Arsenic removal technologies
- › Educating rural populations on the significance of water source testing and the risks associated with arsenic contamination.

Management of Solid and Plastic Waste for reducing contamination of water bodies

- › Establish efficient methods for the collection and segregation of garbage in both urban and rural areas.
- › Implement stringent laws prohibiting single-use plastics and encouraging the use of reusable materials and biodegradable bags as substitutes.

Effectiveness of regulatory laws on prevention of water pollution

Bangladesh has rules to avoid water pollution as well. In order to tackle contamination of the environment, Bangladesh originally created the Water contamination Ordinance 1970. This legislation was later repealed in 1977 by the Environmental Pollution Control legislation. Later on, this country established several laws and regulations to combat water pollution. But their efficacy frequently hinges on appropriate execution, enforcement, and awareness among the public. The weak legal framework, its poor implementation, and ineffective steps and reluctance of the authorities concerned in water resource management are the main reasons behind the failure of water pollution prevention in the country. Bangladesh was the first country to ban polythene, but due to a lack of management strategy and poor rule and regulation execution, it is now the 10th worst country in the world for handling plastic trash. So, the proper implementation of regulatory rules is required to the management of water pollution.” [\[24-29\]](#)

4. Conclusions:

Bangladesh's water pollution problem is urgent and seriously endangers ecosystems, the economy, and public health. The main causes of pollution in rivers, groundwater, and other water sources include plastic trash, agricultural runoff, industrial effluents, and inadequate wastewater treatment. Water is polluted by numerous factors among them industrial wastes are the most prominent. Every industry should have specialized industrial waste treatment facilities to handle its waste materials. These issues are exacerbated by rapid urbanization, insufficient waste management systems, and lax enforcement of environmental regulations. Bangladesh's water resources are essential to both its growth and the welfare of its citizens. Not only are these resources vital to the ecology, but they are also ethically needed to ensure a healthier, more sustainable future for coming generations. By taking immediate and concerted action, the country can overcome the challenges of water pollution and secure a brighter, cleaner future.

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